ModelIt!

Student Activity Pack

Can Nuclear Fusion Power the World?

Modeling Plasma Physics, Energy Balance, and Tokamak Engineering Challenges

Grade 12L3 | NGSS HS-PS1-8 | Energy and Matter

STUDENT NAME _____ DATE _____

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1 Pre-Assessment

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What Do You Know? Answer each question below to the best of your ability. There are no wrong answers — this helps your teacher understand what you already know!

1. **What is the difference between nuclear fission (which powers current nuclear plants) and nuclear fusion (which powers the Sun)?**

2. **Why do you think it has been so difficult to build a fusion power plant even though we have understood the physics for over 70 years?**

3. **What would it mean for the world if we could generate unlimited clean energy from fusion — how would it change energy systems, climate change, and geopolitics?**

4. **What does E = mc2 have to do with nuclear fusion, and why does fusion release so much more energy than chemical reactions like burning fossil fuels?**

2 Scoring Rubric

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Your teacher will use this rubric to assess your work. Read through each level to understand what is expected!

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

3 Component Analysis

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What is a Component?

A **component** is a part of a system that plays a specific role. Components can be **EXTERNAL** (come from outside the system) or **INTERNAL** (happen within the system).

Sort These Components

Deuterium-Tritium Fuel Supply | Magnetic Field Strength | Energy Input (Heating) | Plasma Temperature | Plasma Density | Confinement Time | Fusion Reaction Rate | Neutron Flux on Walls | Wall Material Degradation | Net Energy Output (Q Factor)

EXTERNAL	INTERNAL

? Explain your reasoning: Why did you sort each component the way you did?

4 Model Recording Page

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Draw your model showing all components and their relationships. Use the relationship key below to label your arrows!

+ **POSITIVE:** When A increases, B also increases

− **NEGATIVE:** When A increases, B decreases

Draw your model here

? Explain ONE relationship in your model. What happens when one component changes?

5 Simulation Observations

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Run each scenario in ModelIt! First, write your **prediction**. Then run the simulation and record what you **observe**.

1 Scenario 1: Current Performance

Plasma Temp: 150M degrees C | Density: Moderate | Confinement: 5 seconds | Q = 0.33

MY PREDICTION

WHAT I OBSERVED

2 Scenario 2: ITER Target

Plasma Temp: 150M degrees C | Density: High | Confinement: 500 sec | Energy Input: 50 MW | Target Q = 10

MY PREDICTION

WHAT I OBSERVED

3 Scenario 3: Commercial Plant

Sustained Q > 30 | Continuous operation | Tritium self-sufficiency | Electricity on grid

MY PREDICTION

WHAT I OBSERVED

6 Research & Extend

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Real systems are more complex than our model! Here are additional components scientists study. Choose one and explain how it could connect to our model.

Possible New Components

Plasma Instability Modes — The various ways plasma can become unstable and lose confinement — including kink modes (the plasma column twists), ballooning modes (the plasma bulges outward), and edge-localized modes (ELMs, periodic bursts of hot plasma that erode wall materials) — each requiring specific magnetic field shaping and feedback control to suppress

Superconducting Magnet Cooling — The cryogenic system required to maintain superconducting magnets at operating temperature (4 Kelvin for low-temperature superconductors, 20-77 Kelvin for high-temperature superconductors) while adjacent plasma reaches 150 million degrees — the thermal gradient across a fusion reactor wall is the most extreme in any engineering system

Blanket Module Engineering — The lithium-containing modules that surround the plasma chamber and serve three simultaneous functions: converting fusion neutron kinetic energy to heat for electricity generation, breeding tritium from lithium to refuel the reactor, and shielding the superconducting magnets from neutron damage

My Extension

A Which new component would you add to the model?

B How would it connect to the existing components? (Draw or describe)

C What evidence or reasoning supports your connection?

7 Reflection Questions

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The Big Question

The Sun has powered itself by nuclear fusion for 4.6 billion years, converting hydrogen into helium and releasing enough energy to sustain all life on Earth. If we could replicate this process in a reactor, we would have virtually unlimited clean energy with no carbon emissions and minimal radioactive waste. Why has fusion been 30 years away for the last 70 years, and are we finally on the verge of a breakthrough? Use evidence from your simulation to support your answers.

1. **How does your model illustrate why the Lawson criterion makes fusion so much harder to achieve than fission, which was mastered 80 years ago?**

2. **What does the neutron damage scenario reveal about why achieving net energy gain (Q > 1) is necessary but not sufficient for commercial fusion power?**

3. **Why is tritium self-sufficiency a critical constraint, and what does your model suggest about the minimum tritium breeding ratio required?**

4. **How would improvements in high-temperature superconducting magnets change your model's prediction for the feasibility and timeline of commercial fusion?**

5. **What are the limitations of modeling a fusion reactor with ten components when the real system involves plasma physics, materials science, superconducting magnets, and tritium chemistry simultaneously?**

8 STEM Challenge: Design a Commercial Fusion Power Plant

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YOUR ENGINEERING MISSION

Design the engineering specifications for a commercial fusion power plant that can generate 1 GW of electricity continuously, specifying the plasma parameters, magnetic confinement approach, tritium breeding strategy, wall material solution, and cost targets needed for commercial viability.

THE SCENARIO

A national energy agency has commissioned your engineering team to design the specifications for the first commercial fusion power plant, scheduled for operation by 2045. The plant must generate 1 GW of electricity (enough for 750,000 homes), operate continuously for at least 11 months per year, breed its own tritium fuel, manage neutron damage to reactor walls, and produce electricity at a cost competitive with other clean energy sources. Your model data defines the physics requirements — your engineering creativity must solve the practical challenges.

Engineering Design Thinking

1. WHAT ARE THE PLASMA PARAMETERS (TEMPERATURE, DENSITY, CONFINEMENT TIME, Q FACTOR) YOUR PLANT REQUIRES AND HOW DO THEY COMPARE TO ITER'S TARGETS?

2. WHAT MAGNETIC CONFINEMENT APPROACH DOES YOUR DESIGN USE AND WHAT SUPERCONDUCTING MAGNET TECHNOLOGY DOES IT REQUIRE?

3. HOW DOES YOUR DESIGN HANDLE THE NEUTRON DAMAGE PROBLEM — WHAT WALL MATERIALS AND REPLACEMENT SCHEDULE DOES IT SPECIFY?

4. WHAT IS YOUR ESTIMATED COST PER KILOWATT-HOUR AND HOW DOES IT COMPARE TO SOLAR, WIND, AND FISSION POWER?

My Design

DESIGN AREA — DRAW YOUR SOLUTION

N

! Explain your design using scientific evidence from the simulation:

Real Career Connection

Plasma physicists who study fusion plasma behavior earn $80,000-$160,000/year at national laboratories like PPPL and LLNL. Fusion engineers who design reactor components earn $90,000-$170,000/year at companies like Commonwealth Fusion Systems and TAE Technologies. Superconducting magnet engineers who develop high-field magnets for fusion earn $85,000-$155,000/year. Nuclear materials scientists who study radiation damage to reactor components earn $75,000-$140,000/year.